Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Newton's Second Law — ANSWERS

1 · The Second Law

The rule

Newton's Second Law says that the acceleration of an object is proportional to the resultant force on it, and inversely proportional to its mass.

In plain words: a bigger resultant force gives a bigger acceleration. The same force on a bigger mass gives a smaller acceleration. The acceleration is in the direction of the resultant force.

$F = m\,a$

- F → **resultant force** (N)
- m → **mass** (kg)
- a → **acceleration** (m/s²)

To find the force: $F = m \times a$

To find the acceleration: $a = F \div m$

To find the mass: $m = F \div a$

A resultant force of **1 N** gives a mass of **1 kg** an acceleration of **1 m/s²**.

Mass goes in kilograms. Divide grams by 1000: $250\text{ g} = 250 \div 1000 = 0.25\text{ kg}$.

Task 1 · See the pattern

Use $a = F \div m$ to fill in each table. The first row of each is done for you.

Same mass, bigger force

Force / N	Mass / kg	Acceleration / m/s²
2	2	1
4	2	**2**
8	2	**4**

Same force, bigger mass

Force / N	Mass / kg	Acceleration / m/s²
12	1	12
12	2	**6**
12	4	**3**

When the force doubles, the acceleration **doubles**. When the mass doubles, the acceleration **halves**. That is what "proportional to the force, inversely proportional to the mass" means.

Worked example — copy this shape

Find the force. A 4 kg trolley accelerates at 3 m/s². Calculate the resultant force on it.

Equation $F = m \quad a$

Substitute $F = 4 \times 3$

Calculate $F =$ **12 N**

Find the acceleration. A resultant force of 3000 N acts on a 1200 kg car. Calculate its acceleration.

Equation $F = m \quad a$

Substitute $3000 = 1200 \times a$

Rearrange $a = 3000 \div 1200$

Calculate $a =$ **2.5 m/s²**

What just happened. Each number went in under its own letter. The unknown stayed a letter until the rearrange step, and the answer carries its unit.

Check it. Put your answer back into $F = m \times a$: $1200 \times 2.5 = 3000$ N, the force you started with.

Task 2 · Use $F = ma$

Fill in each gap. Use the shape of the worked example, and give every answer with its unit. Rows 7 and 8 start in grams.

#	Object	Mass	Acceleration	Resultant force
1	a trolley	2 kg	3 m/s^2	**6 N**
2	a cyclist and bike	80 kg	1.5 m/s^2	**120 N**
3	a football being kicked	0.4 kg	**50 m/s^2**	20 N
4	a lorry	12 000 kg	**0.5 m/s^2**	6000 N
5	a motorbike and rider	**300 kg**	2.5 m/s^2	750 N
6	a sprinter leaving the blocks	**75 kg**	5 m/s^2	375 N
7	a falling apple	150 g	9.8 m/s^2	**1.47 N**
8	a toy car	250 g	**2 m/s^2**	0.5 N

2 · Resultant force first

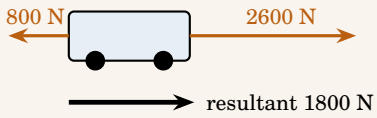
The rule

The F in $F = ma$ is the **resultant** force. When several forces act along a line, find the resultant first: forces in the **same** direction **add**; forces in **opposite** directions **subtract**, and the resultant points the way of the bigger side. Then use $F = ma$.

A resultant force against the motion slows the object down: a **deceleration**.

Worked example — copy this shape

A 1200 kg car. The engine gives a driving force of 2600 N forwards. Air resistance and friction add up to 800 N backwards. Calculate the car's acceleration.

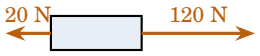


Resultant	$2600 - 800 = 1800$ N forwards
Equation	$F = m \quad a$
Substitute	$1800 = 1200 \times a$
Rearrange	$a = 1800 \div 1200$
Calculate	$a =$ **1.5 m/s^2 forwards**

Check it. The resultant is smaller than the biggest single force, and it points the same way as the bigger side. Your acceleration points that way too.

Task 3 · Find the resultant, then the acceleration

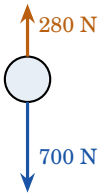
(a) A sledge of mass 50 kg is pulled forwards with a force of 120 N. Friction on the sledge is 20 N. Calculate its acceleration.



Resultant $= 120 - 20 = 100$ N forwards. $a = F \div m = 100 \div 50 =$ **2 m/s^2** forwards.

[3]

(b) A skydiver of mass 70 kg is falling. Her weight is 700 N downwards. Air resistance on her is 280 N upwards. Calculate her acceleration, and give its direction.



Resultant $= 700 - 280 = 420$ N downwards. $a = 420 \div 70 =$ **6 m/s^2 downwards**: she is still speeding up.

[3]

(c) A car of mass 1500 kg is moving forwards. The driver brakes. The only horizontal force on the car is now a braking force of 6000 N backwards.

(i) Calculate the car's acceleration.

(ii) What happens to the car's speed? Use the direction of the resultant force.

(i) $a = 6000 \div 1500 =$ **4 m/s^2 backwards** (a deceleration of 4 m/s^2). (ii) The resultant force is against the motion, so the car slows down.

[3]

(d) A car of mass 900 kg accelerates forwards at 2 m/s^2. Resistive forces of 600 N act on it. Calculate the driving force from the engine.

Use $F = ma$ to find the resultant force first. The driving force has to give that resultant and cancel the 600 N.

Resultant $F = ma = 900 \times 2 = 1800$ N. Driving force $= 1800 + 600 =$ **2400 N** forwards.

[3]

Stretch · Find the acceleration first

No acceleration given? Write the Givens, find a with an equation from Forces 1, then use $F = ma$. Use the size of a; its sign tells you the direction.

(e) A car of mass 1000 kg speeds up from rest to 20 m/s in 8 s. Calculate the resultant force on the car.

Givens

- $s =$ —
- $u = 0$ m/s
- $v = 20$ m/s
- $a =$?
- $t = 8$ s

$a = (v - u) \div t = (20 - 0) \div 8 = 2.5$ m/s^2.
$F = ma = 1000 \times 2.5 =$ **2500 N** forwards.

[3]

(f) A cyclist and bike have a total mass of 80 kg. They brake from 10 m/s to a stop in a distance of 20 m. Calculate the braking force.

Givens

$s = 20$ m
$u = 10$ m/s
$v = 0$ m/s
$a = ?$
$t = —$

$v^2 - u^2 = 2as$: $0^2 - 10^2 = 2 \times a \times 20$, so $a = -100 \div 40 = -2.5$ m/s^2.
$F = ma = 80 \times 2.5 =$ **200 N**, backwards.

[4]

(g) The symbol $\sim$ means "roughly". A family car has a mass of $\sim$1500 kg. In an emergency stop it slows from $\sim$30 m/s to rest in $\sim$3 s. Estimate the braking force.

$a \sim 30 \div 3 = 10$ m/s^2 (a deceleration). $F \sim 1500 \times 10 = \sim$ **15 000 N**, backwards.

[3]